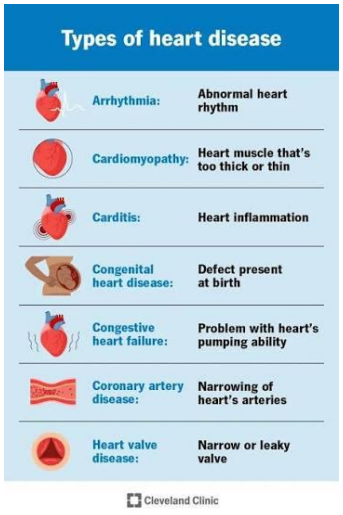
Supervised Learning - Logistics Regression.

Predicting the risk of dying of heart disease.

The World Health Organization estimates 12 million people die annually due to heart disease. This model identifies the most significant heart disease risk variables and forecasts the overall risk.

Model	Accuracy (R^2)
Baseline	84.77%
Logistics Regression	86.02%





```
# -*- coding: utf-8 -*-
"""
Created on Tue Oct 27 12:58:03 2026

@author: Valentin
"""

# WHO estimates that 12 M people die annually due to heart dis
# Part I Logistic Regression

import pandas as pd
framingham = pd.read_csv("framingham.csv")
framingham.head() # Preview the data.

framingham["TenYearCHD"].hist() # View the proportion of cases
```

```
In [6]: %runfile 'C:/Users/Valentin/Desktop/Projects/My Work Projects/AI/01 Regression Logistics and Linear (Supervised Learning).py' --wdir
c:\users\valentin\desktop\projects\my work projects\ai\01 regression logistics and linea
(supervised learning).py:21: UserWarning: To output multiple subplots, the figure
containing the passed axes is being cleared.
  framingham.hist(ax=ax)
                 Generalized Linear Model Regression Results
==============================================================================
Dep. Variable:              TenYearCHD   No. Observations:                 2377
Model:                             GLM   Df Residuals:                     2361
Model Family:                Binomial   Df Model:                           15
Link Function:                   Logit   Scale:                          1.0000
Method:                           IRLS   Log-Likelihood:                -906.78
Date:                 Tue, 27 Oct 2026   Deviance:                       1813.6
Time:                         22:15:42   Pearson chi2:                 2.36e+03
No. Iterations:                      5   Pseudo R-squ. (CS):           0.09035
Covariance Type:             nonrobust
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
Intercept     -6.9052      0.869     -7.947      0.000      -8.608      -5.202
male           0.5545      0.132      4.189      0.000       0.295       0.814
age            0.0557      0.008      6.889      0.000       0.040       0.072
education     -0.0813      0.062     -1.304      0.192      -0.203       0.041
currentSmoker  0.0357      0.194      0.184      0.854      -0.345       0.416
```

Supervised Learning - Linear Regression.

Predicting the price of Wine.

Wine prices were thought to be predicted well only by experts until Princeton professor Orley Ashenfelter proved otherwise. This model predicts wine prices.

Model	Accuracy (Adjusted R^2)
Multiple linear regression	79.4%




```
print (LinearRegModel.summary())
print(wine.corr())

wineTest = pd.read_csv("wineTest.csv")
print(wineTest.info())
# Predict
predictTest = LinearRegModel.predict(sm.ad

import numpy as np
# Compute R-Squared on the test
SSE = np.sum((predictTest - wineTest["Pric
SST = np.sum((wine["Price"].mean() - wineT
r2 = 1 - SSE / SST
```

```
6     763
24    763
Name: WinterRain, dtype: int64
    Year   Price  WinterRain     AGST  HarvestRain  Age
15  1969  6.3459         575  16.5500          244   14
19  1973  6.6367         376  17.0667          123   10
                            OLS Regression Results
==============================================================================
Dep. Variable:                  Price   R-squared:                       0.829
Model:                            OLS   Adj. R-squared:                  0.794
Method:                 Least Squares   F-statistic:                     24.17
Date:                Tue, 27 Oct 2026   Prob (F-statistic):           2.04e-07
Time:                        22:19:45   Log-Likelihood:                -2.1622
No. Observations:                  25   AIC:                             14.32
Df Residuals:                      20   BIC:                             20.42
Df Model:                           4
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
const         -3.4300      1.766     -1.942      0.066      -7.114       0.254
AGST           0.6072      0.099      6.152      0.000       0.401       0.813
WinterRain     0.0011      0.001      2.120      0.047    1.73e-05       0.002
HarvestRain   -0.0040      0.001     -4.652      0.000      -0.006      -0.002
Age            0.0239      0.008      2.956      0.008       0.007       0.041
==============================================================================
```